

Theorem 4.10: (**Chebyshev's Theorem**) The probability that any random variable X will assume a value within k standard deviations of the mean is at least $1 - 1/k^2$. That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

For $k = 2$, the theorem states that the random variable X has a probability of at least $1 - 1/2^2 = 3/4$ of falling within two standard deviations of the mean. That is, three-fourths or more of the observations of any distribution lie in the interval $\mu \pm 2\sigma$. Similarly, the theorem says that at least eight-ninths of the observations of any distribution fall in the interval $\mu \pm 3\sigma$.

Example 4.27: A random variable X has a mean $\mu = 8$, a variance $\sigma^2 = 9$, and an unknown probability distribution. Find

(a) $P(-4 < X < 20)$,

(b) $P(|X - 8| \geq 6)$.

Solution: (a) $P(-4 < X < 20) = P[8 - (4)(3) < X < 8 + (4)(3)] \geq \frac{15}{16}$.

(b) $P(|X - 8| \geq 6) = 1 - P(|X - 8| < 6) = 1 - P(-6 < X - 8 < 6)$

$$= 1 - P[8 - (2)(3) < X < 8 + (2)(3)] \leq \frac{1}{4}.$$

Chebyshev's theorem holds for any distribution of observations, and for this reason the results are usually weak. The value given by the theorem is a lower bound only. That is, we know that the probability of a random variable falling within two standard deviations of the mean can be *no less* than $3/4$, but we never know how much more it might actually be. Only when the probability distribution is known can we determine exact probabilities. For this reason we call the theorem a *distribution-free* result. When specific distributions are assumed, as in future chapters, the results will be less conservative. The use of Chebyshev's theorem is relegated to situations where the form of the distribution is unknown.

Exercises

4.53 Referring to Exercise 4.35 on page 127, find the mean and variance of the discrete random variable $Z = 3X - 2$, when X represents the number of errors per 100 lines of code.

4.54 Using Theorem 4.5 and Corollary 4.6, find the mean and variance of the random variable $Z = 5X + 3$, where X has the probability distribution of Exercise 4.36 on page 127.

4.55 Suppose that a grocery store purchases 5 cartons of skim milk at the wholesale price of \$1.20 per carton and retails the milk at \$1.65 per carton. After the expiration date, the unsold milk is removed from the shelf and the grocer receives a credit from the dis-

tributor equal to three-fourths of the wholesale price. If the probability distribution of the random variable X , the number of cartons that are sold from this lot, is

x	0	1	2	3	4	5
$f(x)$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{2}{15}$	$\frac{3}{15}$	$\frac{4}{15}$	$\frac{3}{15}$

find the expected profit.

4.56 Repeat Exercise 4.43 on page 127 by applying Theorem 4.5 and Corollary 4.6.

4.57 Let X be a random variable with the following probability distribution:

x	-3	6	9
$f(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

(b) It is of interest to know something about the proportion of $Z = X + Y$, the sum of the two proportions. Find $E(X + Y)$. Also find $E(XY)$.

(c) Find $\text{Var}(X)$, $\text{Var}(Y)$, and $\text{Cov}(X, Y)$.

(d) Find $\text{Var}(X + Y)$.

4.71 The length of time Y , in minutes, required to generate a human reflex to tear gas has the density function

$$f(y) = \begin{cases} \frac{1}{4}e^{-y/4}, & 0 \leq y < \infty, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) What is the mean time to reflex?

(b) Find $E(Y^2)$ and $\text{Var}(Y)$.

4.72 A manufacturing company has developed a machine for cleaning carpet that is fuel-efficient because it delivers carpet cleaner so rapidly. Of interest is a random variable Y , the amount in gallons per minute delivered. It is known that the density function is given by

$$f(y) = \begin{cases} 1, & 7 \leq y \leq 8, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Sketch the density function.

(b) Give $E(Y)$, $E(Y^2)$, and $\text{Var}(Y)$.

4.73 For the situation in Exercise 4.72, compute $E(e^Y)$ using Theorem 4.1, that is, by using

$$E(e^Y) = \int_7^8 e^y f(y) dy.$$

Then compute $E(e^Y)$ not by using $f(y)$, but rather by using the second-order adjustment to the first-order approximation of $E(e^Y)$. Comment.

4.74 Consider again the situation of Exercise 4.72. It is required to find $\text{Var}(e^Y)$. Use Theorems 4.2 and 4.3 and define $Z = e^Y$. Thus, use the conditions of Exercise 4.73 to find

$$\text{Var}(Z) = E(Z^2) - [E(Z)]^2.$$

Then do it not by using $f(y)$, but rather by using the first-order Taylor series approximation to $\text{Var}(e^Y)$. Comment!

4.75 An electrical firm manufactures a 100-watt light bulb, which, according to specifications written on the package, has a mean life of 900 hours with a standard deviation of 50 hours. At most, what percentage of the bulbs fail to last even 700 hours? Assume that the distribution is symmetric about the mean.

4.76 Seventy new jobs are opening up at an automobile manufacturing plant, and 1000 applicants show up for the 70 positions. To select the best 70 from among the applicants, the company gives a test that covers mechanical skill, manual dexterity, and mathematical ability. The mean grade on this test turns out to be 60, and the scores have a standard deviation of 6. Can a person who scores 84 count on getting one of the jobs? [Hint: Use Chebyshev's theorem.] Assume that the distribution is symmetric about the mean.

4.77 A random variable X has a mean $\mu = 10$ and a variance $\sigma^2 = 4$. Using Chebyshev's theorem, find

(a) $P(|X - 10| \geq 3)$;

(b) $P(|X - 10| < 3)$;

(c) $P(5 < X < 15)$;

(d) the value of the constant c such that

$$P(|X - 10| \geq c) \leq 0.04.$$

4.78 Compute $P(\mu - 2\sigma < X < \mu + 2\sigma)$, where X has the density function

$$f(x) = \begin{cases} 6x(1-x), & 0 < x < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

and compare with the result given in Chebyshev's theorem.

Review Exercises

4.79 Prove Chebyshev's theorem.

4.80 Find the covariance of random variables X and Y having the joint probability density function

$$f(x, y) = \begin{cases} x + y, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

4.81 Referring to the random variables whose joint probability density function is given in Exercise 3.47 on page 105, find the average amount of kerosene left in the tank at the end of the day.

4.82 Assume the length X , in minutes, of a particular type of telephone conversation is a random variable

Multinomial Experiments and the Multinomial Distribution

The binomial experiment becomes a **multinomial experiment** if we let each trial have more than two possible outcomes. The classification of a manufactured product as being light, heavy, or acceptable and the recording of accidents at a certain intersection according to the day of the week constitute multinomial experiments. The drawing of a card from a deck *with replacement* is also a multinomial experiment if the 4 suits are the outcomes of interest.

In general, if a given trial can result in any one of k possible outcomes $E_1, E_2, \dots, E_k$ with probabilities $p_1, p_2, \dots, p_k$, then the **multinomial distribution** will give the probability that E_1 occurs x_1 times, E_2 occurs x_2 times, $\dots$, and E_k occurs x_k times in n independent trials, where

$$x_1 + x_2 + \dots + x_k = n.$$

We shall denote this joint probability distribution by

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n).$$

Clearly, $p_1 + p_2 + \dots + p_k = 1$, since the result of each trial must be one of the k possible outcomes.

To derive the general formula, we proceed as in the binomial case. Since the trials are independent, any specified order yielding x_1 outcomes for E_1 , x_2 for $E_2, \dots, x_k$ for E_k will occur with probability $p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$. The total number of orders yielding similar outcomes for the n trials is equal to the number of partitions of n items into k groups with x_1 in the first group, x_2 in the second group, $\dots$, and x_k in the k th group. This can be done in

$$\binom{n}{x_1, x_2, \dots, x_k} = \frac{n!}{x_1! x_2! \dots x_k!}$$

ways. Since all the partitions are mutually exclusive and occur with equal probability, we obtain the multinomial distribution by multiplying the probability for a specified order by the total number of partitions.

Multinomial Distribution

If a given trial can result in the k outcomes $E_1, E_2, \dots, E_k$ with probabilities $p_1, p_2, \dots, p_k$, then the probability distribution of the random variables $X_1, X_2, \dots, X_k$, representing the number of occurrences for $E_1, E_2, \dots, E_k$ in n independent trials, is

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k},$$

with

$$\sum_{i=1}^k x_i = n \text{ and } \sum_{i=1}^k p_i = 1.$$

The multinomial distribution derives its name from the fact that the terms of the multinomial expansion of $(p_1 + p_2 + \dots + p_k)^n$ correspond to all the possible values of $f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n)$.

Example 5.7: The complexity of arrivals and departures of planes at an airport is such that computer simulation is often used to model the “ideal” conditions. For a certain airport with three runways, it is known that in the ideal setting the following are the probabilities that the individual runways are accessed by a randomly arriving commercial jet:

Runway 1: $p_1 = 2/9$,
 Runway 2: $p_2 = 1/6$,
 Runway 3: $p_3 = 11/18$.

What is the probability that 6 randomly arriving airplanes are distributed in the following fashion?

Runway 1: 2 airplanes,
 Runway 2: 1 airplane,
 Runway 3: 3 airplanes

Solution: Using the multinomial distribution, we have

$$\begin{aligned} f\left(2, 1, 3; \frac{2}{9}, \frac{1}{6}, \frac{11}{18}, 6\right) &= \binom{6}{2, 1, 3} \left(\frac{2}{9}\right)^2 \left(\frac{1}{6}\right)^1 \left(\frac{11}{18}\right)^3 \\ &= \frac{6!}{2!1!3!} \cdot \frac{2^2}{9^2} \cdot \frac{1}{6} \cdot \frac{11^3}{18^3} = 0.1127. \end{aligned}$$

Exercises

5.1 A random variable X that assumes the values $x_1, x_2, \dots, x_k$ is called a discrete uniform random variable if its probability mass function is $f(x) = \frac{1}{k}$ for all of $x_1, x_2, \dots, x_k$ and 0 otherwise. Find the mean and variance of X .

5.2 Twelve people are given two identical speakers, which they are asked to listen to for differences, if any. Suppose that these people answer simply by guessing. Find the probability that three people claim to have heard a difference between the two speakers.

5.3 An employee is selected from a staff of 10 to supervise a certain project by selecting a tag at random from a box containing 10 tags numbered from 1 to 10. Find the formula for the probability distribution of X representing the number on the tag that is drawn. What is the probability that the number drawn is less than 4?

5.4 In a certain city district, the need for money to buy drugs is stated as the reason for 75% of all thefts. Find the probability that among the next 5 theft cases reported in this district,

- (a) exactly 2 resulted from the need for money to buy drugs;
- (b) at most 3 resulted from the need for money to buy drugs.

5.5 According to *Chemical Engineering Progress* (November 1990), approximately 30% of all pipework failures in chemical plants are caused by operator error.

- (a) What is the probability that out of the next 20 pipework failures at least 10 are due to operator error?
- (b) What is the probability that no more than 4 out of 20 such failures are due to operator error?
- (c) Suppose, for a particular plant, that out of the random sample of 20 such failures, exactly 5 are due to operator error. Do you feel that the 30% figure stated above applies to this plant? Comment.

5.6 According to a survey by the Administrative Management Society, one-half of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. Find the probability that among 6 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is

- (a) anywhere from 2 to 5;
- (b) fewer than 3.

5.7 One prominent physician claims that 70% of those with lung cancer are chain smokers. If his assertion is correct,

- (a) find the probability that of 10 such patients

recently admitted to a hospital, fewer than half are chain smokers;

- (b) find the probability that of 20 such patients recently admitted to a hospital, fewer than half are chain smokers.

5.8 According to a study published by a group of University of Massachusetts sociologists, approximately 60% of the Valium users in the state of Massachusetts first took Valium for psychological problems. Find the probability that among the next 8 users from this state who are interviewed,

- (a) exactly 3 began taking Valium for psychological problems;
(b) at least 5 began taking Valium for problems that were not psychological.

5.9 In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that

- (a) from 3 to 6 have blowouts;
(b) fewer than 4 have blowouts;
(c) more than 5 have blowouts.

5.10 A nationwide survey of college seniors by the University of Michigan revealed that almost 70% disapprove of daily pot smoking, according to a report in *Parade*. If 12 seniors are selected at random and asked their opinion, find the probability that the number who disapprove of smoking pot daily is

- (a) anywhere from 7 to 9;
(b) at most 5;
(c) not less than 8.

5.11 The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that exactly 5 of the next 7 patients having this operation survive?

5.12 A traffic control engineer reports that 75% of the vehicles passing through a checkpoint are from within the state. What is the probability that fewer than 4 of the next 9 vehicles are from out of state?

5.13 A national study that examined attitudes about antidepressants revealed that approximately 70% of respondents believe “antidepressants do not really cure anything, they just cover up the real trouble.” According to this study, what is the probability that at least 3 of the next 5 people selected at random will hold this opinion?

5.14 The percentage of wins for the Chicago Bulls basketball team going into the playoffs for the 1996–97 season was 87.7. Round the 87.7 to 90 in order to use Table A.1.

- (a) What is the probability that the Bulls sweep (4-0) the initial best-of-7 playoff series?
(b) What is the probability that the Bulls win the initial best-of-7 playoff series?
(c) What very important assumption is made in answering parts (a) and (b)?

5.15 It is known that 60% of mice inoculated with a serum are protected from a certain disease. If 5 mice are inoculated, find the probability that

- (a) none contracts the disease;
(b) fewer than 2 contract the disease;
(c) more than 3 contract the disease.

5.16 Suppose that airplane engines operate independently and fail with probability equal to 0.4. Assuming that a plane makes a safe flight if at least one-half of its engines run, determine whether a 4-engine plane or a 2-engine plane has the higher probability for a successful flight.

5.17 If X represents the number of people in Exercise 5.13 who believe that antidepressants do not cure but only cover up the real problem, find the mean and variance of X when 5 people are selected at random.

- 5.18** (a) In Exercise 5.9, how many of the 15 trucks would you expect to have blowouts?
(b) What is the variance of the number of blowouts experienced by the 15 trucks? What does that mean?

5.19 As a student drives to school, he encounters a traffic signal. This traffic signal stays green for 35 seconds, yellow for 5 seconds, and red for 60 seconds. Assume that the student goes to school each weekday between 8:00 and 8:30 a.m. Let X_1 be the number of times he encounters a green light, X_2 be the number of times he encounters a yellow light, and X_3 be the number of times he encounters a red light. Find the joint distribution of X_1 , X_2 , and X_3 .

5.20 According to *USA Today* (March 18, 1997), of 4 million workers in the general workforce, 5.8% tested positive for drugs. Of those testing positive, 22.5% were cocaine users and 54.4% marijuana users.

- (a) What is the probability that of 10 workers testing positive, 2 are cocaine users, 5 are marijuana users, and 3 are users of other drugs?
(b) What is the probability that of 10 workers testing positive, all are marijuana users?

(c) What is the probability that of 10 workers testing positive, none is a cocaine user?

5.21 The surface of a circular dart board has a small center circle called the bull's-eye and 20 pie-shaped regions numbered from 1 to 20. Each of the pie-shaped regions is further divided into three parts such that a person throwing a dart that lands in a specific region scores the value of the number, double the number, or triple the number, depending on which of the three parts the dart hits. If a person hits the bull's-eye with probability 0.01, hits a double with probability 0.10, hits a triple with probability 0.05, and misses the dart board with probability 0.02, what is the probability that 7 throws will result in no bull's-eyes, no triples, a double twice, and a complete miss once?

5.22 According to a genetics theory, a certain cross of guinea pigs will result in red, black, and white offspring in the ratio 8:4:4. Find the probability that among 8 offspring, 5 will be red, 2 black, and 1 white.

5.23 The probabilities are 0.4, 0.2, 0.3, and 0.1, respectively, that a delegate to a certain convention arrived by air, bus, automobile, or train. What is the probability that among 9 delegates randomly selected at this convention, 3 arrived by air, 3 arrived by bus, 1 arrived by automobile, and 2 arrived by train?

5.24 A safety engineer claims that only 40% of all workers wear safety helmets when they eat lunch at the workplace. Assuming that this claim is right, find the probability that 4 of 6 workers randomly chosen will be wearing their helmets while having lunch at the workplace.

5.25 Suppose that for a very large shipment of integrated-circuit chips, the probability of failure for any one chip is 0.10. Assuming that the assumptions underlying the binomial distributions are met, find the probability that at most 3 chips fail in a random sample of 20.

5.26 Assuming that 6 in 10 automobile accidents are due mainly to a speed violation, find the probability that among 8 automobile accidents, 6 will be due mainly to a speed violation

- (a) by using the formula for the binomial distribution;
- (b) by using Table A.1.

5.27 If the probability that a fluorescent light has a useful life of at least 800 hours is 0.9, find the probabilities that among 20 such lights

- (a) exactly 18 will have a useful life of at least 800 hours;
- (b) at least 15 will have a useful life of at least 800 hours;
- (c) at least 2 will *not* have a useful life of at least 800 hours.

5.28 A manufacturer knows that on average 20% of the electric toasters produced require repairs within 1 year after they are sold. When 20 toasters are randomly selected, find appropriate numbers x and y such that

- (a) the probability that at least x of them will require repairs is less than 0.5;
- (b) the probability that at least y of them will *not* require repairs is greater than 0.8.

5.3 Hypergeometric Distribution

The simplest way to view the distinction between the binomial distribution of Section 5.2 and the hypergeometric distribution is to note the way the sampling is done. The types of applications for the hypergeometric are very similar to those for the binomial distribution. We are interested in computing probabilities for the number of observations that fall into a particular category. But in the case of the binomial distribution, independence among trials is required. As a result, if that distribution is applied to, say, sampling from a lot of items (deck of cards, batch of production items), the sampling must be done **with replacement** of each item after it is observed. On the other hand, the hypergeometric distribution does not require independence and is based on sampling done **without replacement**.

Applications for the hypergeometric distribution are found in many areas, with heavy use in acceptance sampling, electronic testing, and quality assurance. Obviously, in many of these fields, testing is done at the expense of the item being tested. That is, the item is destroyed and hence cannot be replaced in the sample. Thus, sampling without replacement is necessary. A simple example with playing

Chapter 6

Some Continuous Probability Distributions

6.1 Continuous Uniform Distribution

One of the simplest continuous distributions in all of statistics is the **continuous uniform distribution**. This distribution is characterized by a density function that is “flat,” and thus the probability is uniform in a closed interval, say $[A, B]$. Although applications of the continuous uniform distribution are not as abundant as those for other distributions discussed in this chapter, it is appropriate for the novice to begin this introduction to continuous distributions with the uniform distribution.

Uniform Distribution	The density function of the continuous uniform random variable X on the interval $[A, B]$ is
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$$f(x; A, B) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B, \\ 0, & \text{elsewhere.} \end{cases}$$

The density function forms a rectangle with base $B - A$ and **constant height** $\frac{1}{B-A}$. As a result, the uniform distribution is often called the **rectangular distribution**. Note, however, that the interval may not always be closed: $[A, B]$. It can be (A, B) as well. The density function for a uniform random variable on the interval $[1, 3]$ is shown in Figure 6.1.

Probabilities are simple to calculate for the uniform distribution because of the simple nature of the density function. However, note that the application of this distribution is based on the assumption that the probability of falling in an interval of fixed length within $[A, B]$ is constant.

Example 6.1:	Suppose that a large conference room at a certain company can be reserved for no more than 4 hours. Both long and short conferences occur quite often. In fact, it can be assumed that the length X of a conference has a uniform distribution on the interval $[0, 4]$.
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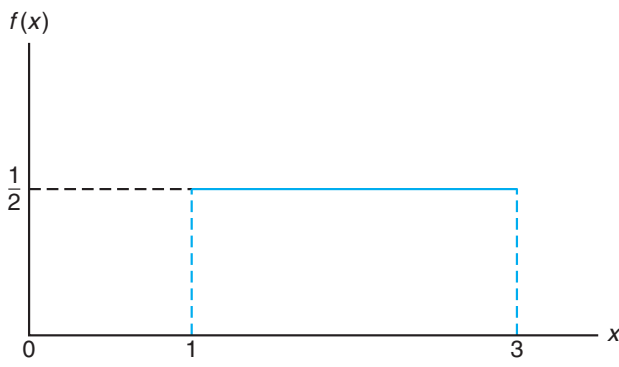


Figure 6.1: The density function for a random variable on the interval $[1, 3]$.

- (a) What is the probability density function?
- (b) What is the probability that any given conference lasts at least 3 hours?

Solution: (a) The appropriate density function for the uniformly distributed random variable X in this situation is

$$f(x) = \begin{cases} \frac{1}{4}, & 0 \leq x \leq 4, \\ 0, & \text{elsewhere.} \end{cases}$$

(b) $P[X \geq 3] = \int_3^4 \frac{1}{4} dx = \frac{1}{4}.$

Theorem 6.1:

The mean and variance of the uniform distribution are

$$\mu = \frac{A + B}{2} \text{ and } \sigma^2 = \frac{(B - A)^2}{12}.$$

The proofs of the theorems are left to the reader. See Exercise 6.1 on page 185.

6.2 Normal Distribution

The most important continuous probability distribution in the entire field of statistics is the **normal distribution**. Its graph, called the **normal curve**, is the bell-shaped curve of Figure 6.2, which approximately describes many phenomena that occur in nature, industry, and research. For example, physical measurements in areas such as meteorological experiments, rainfall studies, and measurements of manufactured parts are often more than adequately explained with a normal distribution. In addition, errors in scientific measurements are extremely well approximated by a normal distribution. In 1733, Abraham DeMoivre developed the mathematical equation of the normal curve. It provided a basis from which much of the theory of inductive statistics is founded. The normal distribution is often referred to as the **Gaussian distribution**, in honor of Karl Friedrich Gauss

Example 6.13: The average grade for an exam is 74, and the standard deviation is 7. If 12% of the class is given As, and the grades are curved to follow a normal distribution, what is the lowest possible A and the highest possible B?

Solution: In this example, we begin with a known area of probability, find the z value, and then determine x from the formula $x = \sigma z + \mu$. An area of 0.12, corresponding to the fraction of students receiving As, is shaded in Figure 6.20. We require a z value that leaves 0.12 of the area to the right and, hence, an area of 0.88 to the left. From Table A.3, $P(Z < 1.18)$ has the closest value to 0.88, so the desired z value is 1.18. Hence,

$$x = (7)(1.18) + 74 = 82.26.$$

Therefore, the lowest A is 83 and the highest B is 82. ┐

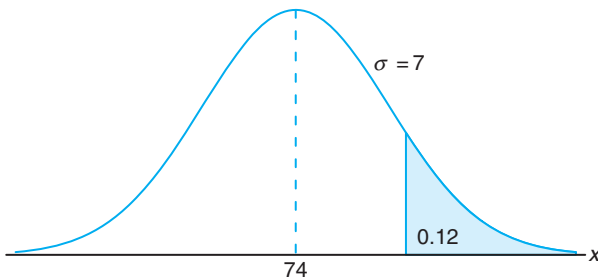


Figure 6.20: Area for Example 6.13.

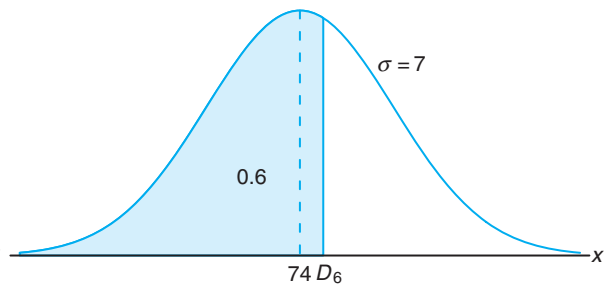


Figure 6.21: Area for Example 6.14.

Example 6.14: Refer to Example 6.13 and find the sixth decile.

Solution: The sixth decile, written D_6 , is the x value that leaves 60% of the area to the left, as shown in Figure 6.21. From Table A.3 we find $P(Z < 0.25) \approx 0.6$, so the desired z value is 0.25. Now $x = (7)(0.25) + 74 = 75.75$. Hence, $D_6 = 75.75$. That is, 60% of the grades are 75 or less. ┐

Exercises

6.1 Given a continuous uniform distribution, show that

- (a) $\mu = \frac{A+B}{2}$ and
- (b) $\sigma^2 = \frac{(B-A)^2}{12}$.

6.2 Suppose X follows a continuous uniform distribution from 1 to 5. Determine the conditional probability $P(X > 2.5 \mid X \leq 4)$.

6.3 The daily amount of coffee, in liters, dispensed by a machine located in an airport lobby is a random

variable X having a continuous uniform distribution with $A = 7$ and $B = 10$. Find the probability that on a given day the amount of coffee dispensed by this machine will be

- (a) at most 8.8 liters;
- (b) more than 7.4 liters but less than 9.5 liters;
- (c) at least 8.5 liters.

6.4 A bus arrives every 10 minutes at a bus stop. It is assumed that the waiting time for a particular individual is a random variable with a continuous uniform distribution.

- (a) What is the probability that the individual waits more than 7 minutes?
- (b) What is the probability that the individual waits between 2 and 7 minutes?

6.5 Given a standard normal distribution, find the area under the curve that lies

- (a) to the left of $z = -1.39$;
- (b) to the right of $z = 1.96$;
- (c) between $z = -2.16$ and $z = -0.65$;
- (d) to the left of $z = 1.43$;
- (e) to the right of $z = -0.89$;
- (f) between $z = -0.48$ and $z = 1.74$.

6.6 Find the value of z if the area under a standard normal curve

- (a) to the right of z is 0.3622;
- (b) to the left of z is 0.1131;
- (c) between 0 and z , with $z > 0$, is 0.4838;
- (d) between $-z$ and z , with $z > 0$, is 0.9500.

6.7 Given a standard normal distribution, find the value of k such that

- (a) $P(Z > k) = 0.2946$;
- (b) $P(Z < k) = 0.0427$;
- (c) $P(-0.93 < Z < k) = 0.7235$.

6.8 Given a normal distribution with $\mu = 30$ and $\sigma = 6$, find

- (a) the normal curve area to the right of $x = 17$;
- (b) the normal curve area to the left of $x = 22$;
- (c) the normal curve area between $x = 32$ and $x = 41$;
- (d) the value of x that has 80% of the normal curve area to the left;
- (e) the two values of x that contain the middle 75% of the normal curve area.

6.9 Given the normally distributed variable X with mean 18 and standard deviation 2.5, find

- (a) $P(X < 15)$;
- (b) the value of k such that $P(X < k) = 0.2236$;
- (c) the value of k such that $P(X > k) = 0.1814$;
- (d) $P(17 < X < 21)$.

6.10 According to Chebyshev's theorem, the probability that any random variable assumes a value within 3 standard deviations of the mean is at least $8/9$. If it is known that the probability distribution of a random variable X is normal with mean μ and variance σ^2 , what is the exact value of $P(\mu - 3\sigma < X < \mu + 3\sigma)$?

6.11 A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,

- (a) what fraction of the cups will contain more than 224 milliliters?
- (b) what is the probability that a cup contains between 191 and 209 milliliters?
- (c) how many cups will probably overflow if 230-milliliter cups are used for the next 1000 drinks?
- (d) below what value do we get the smallest 25% of the drinks?

6.12 The loaves of rye bread distributed to local stores by a certain bakery have an average length of 30 centimeters and a standard deviation of 2 centimeters. Assuming that the lengths are normally distributed, what percentage of the loaves are

- (a) longer than 31.7 centimeters?
- (b) between 29.3 and 33.5 centimeters in length?
- (c) shorter than 25.5 centimeters?

6.13 A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live

- (a) more than 32 months;
- (b) less than 28 months;
- (c) between 37 and 49 months.

6.14 The finished inside diameter of a piston ring is normally distributed with a mean of 10 centimeters and a standard deviation of 0.03 centimeter.

- (a) What proportion of rings will have inside diameters exceeding 10.075 centimeters?
- (b) What is the probability that a piston ring will have an inside diameter between 9.97 and 10.03 centimeters?
- (c) Below what value of inside diameter will 15% of the piston rings fall?

6.15 A lawyer commutes daily from his suburban home to his midtown office. The average time for a one-way trip is 24 minutes, with a standard deviation of 3.8 minutes. Assume the distribution of trip times to be normally distributed.

- (a) What is the probability that a trip will take at least $1/2$ hour?
- (b) If the office opens at 9:00 A.M. and the lawyer leaves his house at 8:45 A.M. daily, what percentage of the time is he late for work?